

MODEL QUESTION PAPER
INTRODUCTION TO MACHINE LEARNING SET-I

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. Define machine Learning with suitable example
2. What do you mean by linear regression?
3. Which are the four types of noise?
4. What is model selection?
5. What is the main drawback of cross validation technique?
6. What are the applications of bayes theorem?
7. What is the geometric intuition behind SVM?
8. What is the advantage of expectation step?
9. Where does the hidden Markov model is used?

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. What is PAC learning model? Why is PAC learning important?
2. What is the basic principle of dimensionality reduction? What are the three ways of reducing dimensionality?
3. What are the objectives of principle component analysis?
4. a)How do we use bootstrapping in machine learning?
b)Does bootstrapping reduce overfitting?
5. a) Explain k class classification with example. How is it different from 2 class classification?
6. Which are the popular algorithms that can be used for multiclass classification?
7. Explain the concepts of training, validation, cross validation and test set?
8. What are the challenges of unsupervised learning? Which types of problems can be solved by unsupervised learning?

9. Explain

a) Decision tree

b) Random forest

10. What is Artificial Neural Network? How it differ from normal computers?

11. Explain any two clustering algorithms.

12. What are the main components of a Markov decision process?

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. What is generalisation? Which issues are to be considered while selecting a model for applying machine learning in a given problem?

OR

2. What are different types of naive Bayes classifiers? How does a Bayesian classifier work?

3. What is k- fold cross validation? Explain with example.

OR

4 How do you measure the performance of a classifier in machine learning?

5. What are the three fundamental problems associated with Hidden Markov Model ?

OR

6. Write down the real world applications and limitations of SVM.

7. Explain the biological prototype of neuron. Also explain the characteristics of neuron.

OR

8.What are the different types of association rules? Write down the applications of association rule mining.

9.Elaborate on the types of machines and discuss the components in the design of a learning system.

OR

10. What is expectation maximization algorithm? How does this work?

11.Where ROC curve is used in machine learning? How do you interpret the ROC curves?

OR

12.How do you calculate maximum likelihood estimation? What assumptions are generally made?

(6x7=42)

(8x3=24)

MODEL QUESTION PAPER (REV-21)
INTROCTION TO MACHINE LEARNING SET-2

(Maximum Marks : 75)

TIME : 2HRS

PART -A

Answer all questions in one word or one sentence

I.

1. What do you understand by machine learning?
2. List the basic Components of learning process?
3. Name the two approaches to Subset Selection?
4. What is feature selection?
5. What do you mean by stratification?
6. What is the use of validation set?
7. Why K-means clustering is used?
8. List any two applications of HMM.?
9. What is artificial neuron?

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. What are different types of machine learning?
2. Distinguish between classification and regression?
3. What do you mean by reinforcement learning?
4. Explain PAC learning with example.
5. Explain Noise in data with example
6. Why dimensionality reduction is useful?
7. Explain the concepts behind model selection and generalization?
8. Explain Cross Validation and resampling method.
9. Describe the “one-against-one” method for learning multiple classes.
10. Write Perceptron learning algorithm?

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. What are the differences between supervised and unsupervised learning?

OR

2. Explain the concept of Association rule learning. Give the names of two algorithms for generating association rules.?

3. What is meant by overfitting and under fitting of data with examples?

OR

4. What is PCA? Also explain steps in PCA?

5. Explain Cross Validation method? What are the problems encountered in cross validation method.?

OR

6. Explain Bootstrapping method for training/validation sample selection. Compare it with K fold cross validation?
7. What is ROC curves? Also explain ROC curves with example?

OR

8. Explain Naïve Bayes Classifier Algorithm and Working of Naïve Bayes' Classifier?
9. Explain basic structure of an artificial neuron with illustration.?

OR

10. Explain the SVM algorithm in detail ?

11. Compare Observable Markov Model and Hidden Markov Model with example?

OR

12. Briefly explain the three problems associated with HMM?

(8x3=24)